1A 1B What kind of data is this?

LEARNING INTENTIONS

- To be able to classify data as categorical or numerical.
- To be able to further classify categorical data as nominal or ordinal.
- To be able to further classify numerical data as discrete or continuous.
- To be able to construct frequency and percentage frequency tables for categorical data.
- To be able to construct bar charts and percentage bar charts from frequency tables.
- To be able to interpret and describe frequency tables and bar charts.
- To be able to answer statistical questions about a categorical variable.

ACTIVATING PRIOR KNOWLEDGE

1. Classify the following data as nominal, ordinal, discrete or continuous:

a The time between people arriving at the coffee shop.

_____

b The number of people in the queue at the coffee shop.

_____

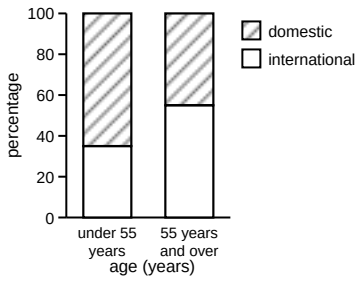
c Customer's coffee preference (latte, cappuccino, black).

_____

d Customer's rating of the coffee (excellent, quite good, not that good).

_____

2. The percentage segmented bar chart provided shows the *age* (under 55 years, 55 years and over) of visitors at a travel convention, segmented by *preferred travel destination* (domestic, international).



The variables *age* (under 55 years, 55 years and over) and *preferred travel destination* (domestic, international) are

A. both categorical variables.
B. both numerical variables.
C. a numerical variable and a categorical variable respectively.
D. a categorical variable and a numerical variable respectively.
E. a discrete variable and a continuous variable respectively.

Guess: **%** of students who got this right _____ Most common wrong answer _____

VCAA 2021 Exam 1 · Data analysis Q1

3. The data in the table shown was collected in a study of the association between the variables *frequency of nightmares* (low, high) and *snores* (no, yes).

frequency of nightmares	***snores***		**total**
	no	**yes**	
low	80	58	138
high	11	12	23
total	**91**	**70**	**161**

The variables in this study, *frequency of nightmares* (low, high) and *snores* (no, yes), are

A. ordinal and nominal respectively.
B. nominal and ordinal respectively.
C. both numerical.
D. both ordinal.
E. both nominal.

Guess: **%** of students who got this right _____ Most common wrong answer _____

VCAA 2020 Exam 1 · Data analysis Q10

Example 1 · Types of data

Classify the following data as nominal, ordinal, discrete or continuous.

a The number of chocolate chips in each of 10 cookies is counted.

b The time taken for 20 students to complete a puzzle is recorded in seconds.

c Member of a football club were asked to rate how they felt about the current coach, 1 = Very satisfied, 2 = Satisfied, 3 = Indifferent, 4 = Dissatisfied, 5 = Very dissatisfied.

d Students are asked to each choose their preferred colour from the list 1 = Blue, 2 = Green, 3 = Red, 4 = Yellow.

e Students weights were classified as ‘less than 60kg’, ‘60kg - 80kg’ or ‘more than 80kg’.

You try 1

Classify the following variables as either nominal, ordinal, discrete or continuous.

a *car brand* (1 = Toyota, 2 = Holden, 3 = Ford, 4 = other)

b *number of employees*

c *weight of textbook* (kg)

d *height of basketball players* (cm)

e *perfume brand*

Example 2 · Frequency table for a categorical variable

A group of 11 preschool children were asked to choose between chocolate and vanilla ice-cream (C = chocolate, V = vanilla):

C V V C C V C C C V V

Construct a frequency table (including percentage frequencies) to display the data.

flavour	**Frequency**	
	Number	**Percentage**
chocolate	6	54.5
vanilla	5	45.5
Total	**11**	**100.0**

You try 2

The students in a prep class were asked the question, ‘Would you describe your teacher’s height as short, average or tall?’. Their responses were as follows:

average	short	tall	tall	short	short
average	average	tall	short	average	short
average	average	average	tall	tall	short
average	tall	tall	average	average	tall

Use this data to create a frequency table displaying both frequency and percentage frequency, correct to the nearest decimal place.

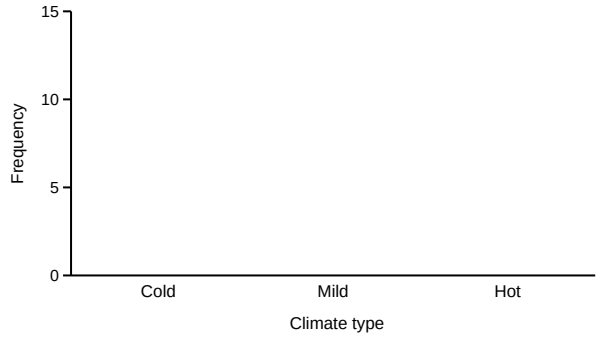
Example 3 · Constructing a bar chart from a frequency table

The climate type of 23 countries is classified as ‘cold’, ‘mild’ or ‘hot’. The results are summarised in the table opposite.

Climate type	Frequency	
	Number	**Percentage**
Cold	3	13.0
Mild	14	60.9
Hot	6	26.1
Total	**23**	**100.0**

a What is the level of measurement of the variable *climate type*?

b Construct a frequency bar chart to display this information.



You try 3

24 students in a prep class were asked the question, ‘Would you describe your teacher’s height as short, average or tall?’. Their responses are recorded in the frequency table shown.

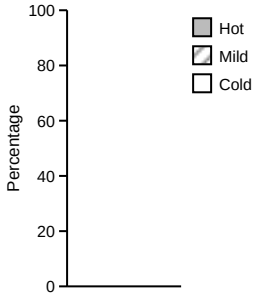
Use the frequency table to construct a frequency bar chart.

teacher’s height	frequency	
	number	**%**
short	6	25.0
average	10	41.7
tall	8	33.3
total	**24**	**100.0**

Example 4 · Constructing a percentage segmented bar chart

The climate type of 23 countries is classified as ‘cold’, ‘mild’ or ‘hot’.
Construct a percentage frequency segmented bar chart to display this information.

Climate type	Frequency	
	Number	**Percentage**
Cold	3	13.0
Mild	14	60.9
Hot	6	26.1
Total	**23**	**100.0**

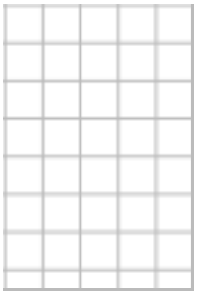


You try 4

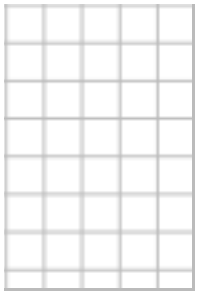
24 students in a prep class were asked the question, ‘Would you describe your teacher’s height as short, average or tall?’. Their responses are recorded in the frequency table shown.

teacher’s height	frequency	
	number	**%**
short	6	25.0
average	10	41.7
tall	8	33.3
total	**24**	**100.0**

a Use the frequency table to construct a segmented bar chart.



b Use the frequency table to construct a percentage segmented bar chart.



EXAMPLE 5 · DESCRIBING THE DISTRIBUTION OF A CATEGORICAL VARIABLE

In an investigation of the variation of climate type across countries, the climate types of 23 countries were classified as ‘cold’, ‘mild’ or ‘hot’. The data are displayed in a frequency table to show the percentages.

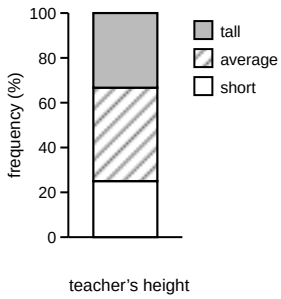
Use the information in the frequency table to write a concise report on the distribution of climate types across these 23 countries.

Climate type	Frequency	
	Number	**Percentage**
Cold	3	13.0
Mild	14	60.9
Hot	6	26.1
Total	**23**	**100.0**

The _____ (variable) of _____ (number) _____ (individuals) were classified as being _____ (categories). The _____ (majority / most) of the _____ (individuals), _____ (%), were found to _____ (modal category). Of the remaining _____ (individuals), _____ (%) were found to _____ (category), while _____ (%) were found to _____ (category).

YOU TRY 5

24 students in a prep class were asked the question, 'Would you describe your teacher's height as short, average or tall?'. Their responses are shown in the given percentage segmented bar chart.



a Find the modal category of the data set.

b Describe the distribution of the data set.

EXIT TASKS

1. Data relating to the following five variables was collected from insects that were caught overnight in a trap: *colour*, *name of species*, *number of wings*, *body length* (in millimetres), *body weight* (in milligrams).

The number of these variables that are discrete variables is

A. 1 **B.** 2 **C.** 3 **D.** 4 **E.** 5

Guess: **%** of students who got this right _____ Most common wrong answer _____

VCAA 2020 Exam 1 · Data analysis Q7

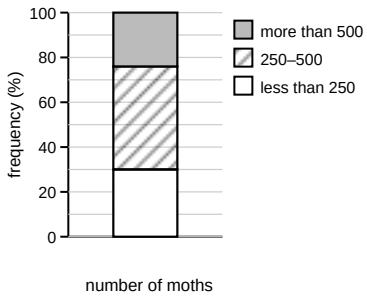
2. The variables *number of moths* (less than 250, 250–500, more than 500) and *trap type* (sugar, scent, light) are

A. both nominal variables.
B. both ordinal variables.
C. a numerical variable and a categorical variable respectively.
D. a nominal variable and an ordinal variable respectively.
E. an ordinal variable and a nominal variable respectively.

Guess: **%** of students who got this right _____ Most common wrong answer _____

VCAA 2017 Exam 1 · Data analysis Q7

3. A study was conducted that investigated the number of moths caught in a sugar moth trap (less than 250, 250–500, more than 500). The results are summarised in the percentage segmented bar chart shown.



There were 300 sugar traps.

The number of sugar traps that caught less than 250 moths is closest to

A. 30 **B.** 90 **C.** 250 **D.** 300 **E.** 500

Guess: **%** of students who got this right _____ Most common wrong answer _____

Adapted from VCAA 2017 Exam 1 · Data analysis Q5

SKILLS CHECKLIST

Tick when you can do it

- ☐ **1. I can identify types of data.**
- ☐ **2. I can construct a frequency table for categorical data.**
- ☐ **3. I can construct a bar chart from a frequency table.**
- ☐ **4. I can construct a percentage segmented bar chart from a frequency table.**
- ☐ **5. I can describe the distribution of a categorical variable.**

HOMEWORK · CAMBRIDGE EXERCISES 1A AND 1B

Tick when done

EXERCISE 1A	EXERCISE 1B	REVISION BOOK
Q1 · Q2 · Q3 Q4–6 *(Exam 1 style)* ☐	Q1–3 · Q4–5 · Q6–8 Q9 *(Exam 1 style)* ☐	1A and 1B sections ☐

Teacher stamp

Teacher signature _____ Date _____

Parent signature _____ Date _____